Enhancing Metaheuristics for Pistachio Supply Chain Optimization

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Abstract

Efficient supply chain management is critical for high-value agricultural products. While mixed-integer linear programming models for the pistachio industry exist, the literature lacks efficient heuristic approaches with clear resolution methodologies. This paper addresses this issue by presenting an enhanced Variable Neighborhood Search (VNS) approach. Unlike prior studies that used real-valued encoding with unspecified decoding steps, we propose a transparent integer encoding scheme and a rigorous decoding algorithm. Furthermore, we develop domain-specific neighborhood structures that significantly improve search convergence compared to classical operators. We benchmark our method against Genetic Algorithms (GA), Iterated Local Search (ILS), and an exact solver. The experiments reveal that while the exact method provide optimal solutions for small instances, the proposed VNS is robust and superior in finding high-quality solutions for large-scale datasets, outperforming other metaheuristics in both stability and objective function values.

Keywords: Supply Chain Optimization, Metaheuristics, Variable Neighborhood Search, Sustainable Logistics

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1. Introduction

The growing role of agriculture in the economic development of many countries highlights the importance of integrating this sector with other industries to stimulate overall economic growth. Agriculture not only supports food security but also acts as a major contributor to exports, generating significant income and strengthening other sectors. Among various agricultural products, pistachios stand out due to their high economic and nutritional value. Major producers such as the US, Turkey, and Iran dominate global pistachio production, remaining the leading producers worldwide in 2022 and 2023 (Statista (2023)). As a result, the strategic importance of pistachios in international trade continues to grow.

With the increasing global demand for pistachios, the need for efficient supply chain networks becomes more critical. Proper supply chain management is vital for maintaining product quality, minimizing environmental impacts, and ensuring the smooth flow of goods from production to consumers. A key challenge in pistachio production is the management of its by-products, such as hulls and shells, which, while free of heavy metals or pathogens, can degrade soil quality and contribute to environmental pollution if not properly handled (Doula et al. (2014); Bartzas and Komnitsas (2017)). Furthermore, improper disposal of these materials may lead to the growth of molds that increase the risk of aflatoxin contamination, posing serious health risks such as liver cancer (Hokmabadi (2018); Dronavalli and Kang (2019)). Therefore, eco-friendly practices for reusing agricultural waste are essential to mitigate both environmental and health hazards.

Moreover, inefficiencies in supply chain networks for food and agricultural products, including pistachios, can have significant implications for product quality and trade (Casper and Sundin (2018)). Addressing these environmental and logistical concerns is crucial to sustaining both the economic and ecological viability of the pistachio industry. In the broader context of sustainable supply chains, several studies have focused on optimizing resource use and minimizing waste through recovery and recycling practices. For example, sustainable supply chain models designed for agriculture and other industries have shown significant environmental benefits by reducing waste and reusing by-products for processes like biogas and compost production (Cheraghalipour and Roghanian (2022)). Such approaches have been successfully applied in sectors such as manufacturing and waste recovery, where efficient resource management and cost reduction have been key drivers (Devika et al. (2014); Zohal and Soleimani (2016)). These strategies underline the importance of sustainability in modern supply chains and the potential for agricultural industries to implement similar measures.

In related research, Rajabi-Kafshgar et al. (Rajabi-Kafshgar et al. (2023)) focused on optimizing the pistachio supply chain using various algorithms, primarily population-based approaches such as Genetic Algorithms (GA) (Fathollahi-Fard and Hajiaghaei-Keshteli (2018); Xie et al. (2022)), among others. However, a significant limitation in their work is the lack of methodological transparency. Specifically, the authors employed a real-valued encoding scheme without detailing the decoding procedure required to transform these values into a feasible transport tree. This omission hinders reproducibility and obscures the mechanism by which constraints are handled. Furthermore, the literature has increasingly questioned the effectiveness of "metaphor-based" metaheuristics, suggesting that established methods often perform equally well or better without the complexity of novel analogies (Campelo and Aranha (2023); Aranha et al. (2022)).

Addressing these gaps, this research focuses on designing an optimal supply chain network for pistachios by leveraging the Variable Neighborhood Search (VNS) algorithm (Gendreau and Potvin (2010)). While we adopt the same mixed-integer linear programming (MILP) model formulation from (Rajabi-Kafshgar et al. (2023)), we propose a robust overhaul of the resolution methodology. We introduce a transparent integer-based priority encoding and provide a step-by-step description of the decoding algorithm, ensuring reproducibility. Furthermore, we develop problem-specific neighborhood operators designed to exploit the specific structure of the chain. These specialized operators are contrasted with classical moves to demonstrate their efficacy in exploring the solution space.

To validate the proposed approach, we conduct a comprehensive comparative analysis against a Genetic Algorithm (GA), an Iterated Local Search (ILS), and an exact Branch-and-Cut method (using the Gurobi solver). Our findings demonstrate that the proposed VNS with specialized operators consistently outperforms the other metaheuristics in terms of solution quality. On top of that, we show that for large-scale instances, where exact methods become infeasible due to memory limitations and excessive runtime, the proposed heuristic offers a vital trade-off, providing high-quality feasible solutions efficiently.

The remainder of this paper is organized as follows. Section 2 presents the problem definition and the mathematical formulation. Section 3 details the heuristic approach, introducing the integer encoding, the specific decoding steps, and the novel domain-specific neighborhood structures. Section 4 reports the computational experiments, offering a statistical comparison between VNS, GA, ILS, and the exact solver across varying instance sizes. Finally, Section 5 summarizes the key findings and suggests future research directions.

2. Problem Definition

The logistics network proposed by Rajabi-Kafshgar et al. (2023), illustrated in Figure 1, describes the flow of products and by-products along the pistachio supply chain. The process begins with the farmers (i), who send the harvested pistachios to the processing centers (j). At these centers, the pistachios are transformed into three main components: open-shell pistachios, raw kernels, and organic waste.

Each of these components follows specific flows. Open-shell pistachios are forwarded to pistachio factories (k), which process and distribute them to final customers ($n1$). Raw kernels are sent to oil extraction centers (e), responsible for producing pistachio oil, which is directed to both final customers ($n2$) and cosmetic factories (s). These, in turn, process the oil into cosmetic products that are supplied to their respective consumers ($n3$). Organic waste generated in the initial processing is sent to composting centers (q), where it is transformed into compost, subsequently distributed to composting customers (m).

Although, in practical terms, the produced compost could be reused by the original farmers, establishing a closed-loop system, the adopted mathematical model does not explicitly represent this feedback. Instead, compost customers are not directly associated with the inital producers, characterizing an implicit feedback cycle. This approach, already adopted in previous works (Pishvaee et al. (2010); Ahmadi and Amin (2019); Banasik et al. (2016)), preserves the structural simplicity of the model. However, the inclusion of rigorous modeling of this return loop represents a promising direction for future investigations, especially in the context of sustainable supply chains.

The mixed linear problem aims to minimize total costs (1), including opening, production, and transportation costs, subject to the following capacity and demand constraints, tolerating no shortages (see more details on variables in Table 5 of Rajabi-Kafshgar et al. (2023)):

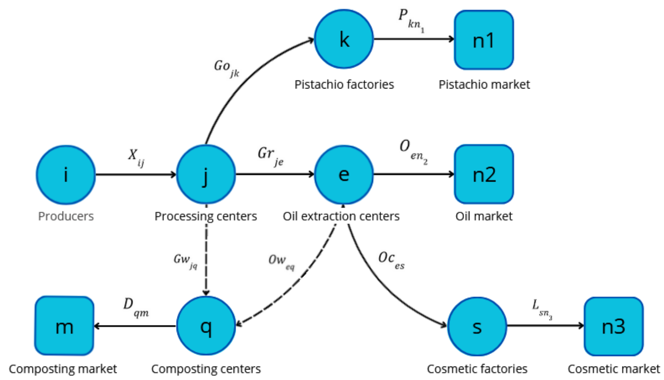

Figure 1: Structure of the proposed pistachio supply chain.

$$\min\ z_1 + z_2 + z_3 \tag{1}$$

$$\text{s.t.:} \quad \sum_{j=1}^{J} X_{ij} \leq Cpa_i, \quad \forall i \in I \tag{2}$$

$$\sum_{i=1}^{I} X_{ij} \leq Cpu_j \times U_j, \quad \forall j \in J \tag{3}$$

$$\sum_{e=1}^{E} Ow_{eq} + \sum_{j=1}^{J} Gw_{jq} \leq Cpy_q \times Y_q, \quad \forall q \in Q \tag{4}$$

$$\sum_{j=1}^{J} Go_{jk} \leq Cpw_k \times W_k, \quad \forall k \in K \tag{5}$$

$$\sum_{j=1}^{J} Gr_{je} \leq Cpr_e \times R_e, \quad \forall e \in E \tag{6}$$

$$\sum_{e=1}^{E} O_{en_2} \leq Cpv_s \times V_s, \quad \forall s \in S \tag{7}$$

$$\sum_{k=1}^{K} Go_{jk} + \sum_{e=1} Gr_{je} + \sum_{q=1}^{Q} Gw_{jq} \leq \sum_{i=1} X_{ij}, \quad \forall j \in J \tag{8}$$

$$\sum_{k=1}^{K} Go_{jk} = (1-\beta) \sum_{i=1}^{I} X_{ij}\theta_1, \quad \forall j \in J \tag{9}$$

$$\sum_{e=1}^{K} Gr_{je} = (1-\beta) \sum_{i=1}^{I} X_{ij}\theta_2, \quad \forall j \in J \tag{10}$$

$$\sum_{q=1}^{Q} Gw_{jq} = \sum_{i=1}^{I} X_{ij}\theta_3, \quad \forall j \in J \tag{11}$$

$$\theta_1 + \theta_2 + \theta_3 = 1 \tag{12}$$

$$\sum_{n_1=1}^{N_1} P_{kn_1} \leq \sum_{j=1}^{J} Go_{jk} \times \gamma k, \quad \forall k \in K \tag{13}$$

$$\sum_{n_2=1}^{N_2} O_{en_2} + \sum_{s=1}^{S} Oc_{es} \leq \sum_{j=1} Gr_{je} \times (1-\lambda), \quad \forall e \in E \tag{14}$$

$$\sum_{q=1}^{Q} Ow_{eq} \leq \sum_{j=1}^{J} Gr_{je} \times \lambda, \quad \forall e \in E \tag{15}$$

$$\sum_{n_3=1}^{N_3} L_{sn_1} \leq \sum_{e=1}^{E} Oc_{es} \times \gamma s, \quad \forall s \in S \tag{16}$$

$$\sum_{m=1}^{M} D_{qm} \leq \left(\sum_{j=1}^{J} Gw_{jq} + \sum_{e=1}^{E} Ow_{eq} \right) \times \gamma q, \quad \forall q \in Q \tag{17}$$

$$\sum_{k=1}^{K} P_{kn_1} \geq Dp_{n_1}, \quad \forall n_1 \in N_1 \tag{18}$$

$$\sum_{e=1}^{E} O_{en_2} \geq Du_{n_2}, \quad \forall n_2 \in N_2 \tag{19}$$

$$\sum_{s=1}^{S} L_{sn_3} \geq Ds_{n_3}, \quad \forall n_3 \in N_3 \tag{20}$$

$$\sum_{q=1}^{Q} D_{qm} \leq Dc_m, \quad \forall m \in M \tag{21}$$

- The objective function (1) minimizes the total costs of the supply chain, including opening, transportation, and production costs. The costs are divided into three components: z_1 (22), z_2 (23), and z_3 (24), which represent facility opening costs, transportation costs, and production costs, respectively, as shown in the following equations:

$$
\begin{aligned}
z_1 = {} & \sum_{j=1}^{J} Fu_j \cdot U_j + \sum_{q=1}^{Q} Fy_q \cdot Y_q + \sum_{k=1}^{K} Fw_k \cdot W_k \\
& + \sum_{e=1}^{E} Fr_e \cdot R_e + \sum_{s=1}^{S} Fv_s \cdot V_s
\end{aligned} \tag{22}
$$

$$
\begin{aligned}
z_2 = {} & \sum_{i=1}^{I} \sum_{j=1}^{J} CI_i \cdot X_{ij} + \sum_{j=1}^{J} \sum_{k=1}^{K} Cu_{1j} \cdot Go_{jk} \\
& + \sum_{j=1}^{J} \sum_{e=1}^{E} Cu_{2j} \cdot Gr_{je} + \sum_{q=1}^{Q} \sum_{m=1}^{M} Cy_q \cdot D_{qm} \\
& + \sum_{k=1}^{K} \sum_{n_1=1}^{N_1} Cw_k \cdot P_{kn_1} \\
& + \sum_{e=1}^{E} Cr_e \cdot \left(\sum_{n_2=1}^{N_2} O_{en_2} + \sum_{s=1}^{S} Oc_{es} \right) \\
& + \sum_{s=1}^{S} \sum_{n_3=1}^{N_3} Cv_s \cdot L_{sn_3}
\end{aligned} \tag{23}
$$

$$
\begin{aligned}
z_3 = {} & \sum_{i=1}^{I} \sum_{j=1}^{J} CX_{ij} \cdot X_{ij} + \sum_{j=1}^{J} \sum_{k=1}^{K} CK_{jk} \cdot Go_{jk} \\
& + \sum_{j=1}^{J} \sum_{q=1}^{Q} CJ_{jq} \cdot Gw_{jq} + \sum_{e=1}^{E} \sum_{s=1}^{S} CS_{es} \cdot Oc_{es} \\
& + \sum_{e=1}^{E} \sum_{n_2=1}^{N_2} CN_{en_2} \cdot Oc_{en_2} + \sum_{e=1}^{E} \sum_{q=1}^{Q} CQ_{eq} \cdot Ow_{eq} \\
& + \sum_{s=1}^{S} \sum_{n_3=1}^{N_3} Cl_{sn_3} \cdot L_{sn_3} + \sum_{k=1}^{K} \sum_{n_1=1}^{N_1} Cp_{kn_1} \cdot P_{kn_1} \\
& + \sum_{q=1}^{Q} \sum_{m=1}^{M} Cd_{qm} \cdot D_{qm}
\end{aligned} \tag{24}
$$

- The variables R_j, Y_q, W_k, Z_e, and V_s are binary and determine whether certain elements of the model are active. For example, $R_j = 1$ indicates that facility j is operational, and $R_j = 0$ otherwise. The other variables in the model are continuous and non-negative, representing real values greater than or equal to zero. These variables model flows, quantities, and other measurable factors within the system.

- Constraints (2)-(7) address the capacity and demand limitations of each facility in the supply chain. In particular, constraint (2) limits the quantity of products transported from farmers to distribution centers. Constraints (3)-(7) ensure that the quantities sent to processing centers, composting centers, pistachio factories, oil extraction centers, and cosmetic factories do not exceed the capacities of these facilities.

- The quantity of products shipped from any facility must not exceed the orders received by that facility. Consequently, constraint (8) establishes that the quantity of products shipped from a processing center must be less than or equal to the quantity received by other facilities. Constraints (9)-(12) define the quantities of open-shell pistachios, raw kernels, and waste generated during the hulling process at each processing center. Similarly, constraints (13)-(17) govern the flows within pistachio factories, oil extraction centers, cosmetic factories, and composting centers.

- Finally, to ensure that customer demand is met, constraints (18)-(21) ensure that the quantities of products delivered to pistachio, pistachio oil, cosmetics, and compost customers satisfy or exceed their respective demands.

3. Methodology

To solve the problem described in Section 2, we propose a metaheuristic algorithm.

3.1. Initial Solution

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3.2. Neighborhood Structures

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4. Computational Experiments

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5. Conclusion

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